

A Complete Six-Vertex Classification for a Chromatic Graphon Lower Bound

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Abstract

For a finite graph H , let

$$\Phi_H(p) = (1-p)^{v(H)} \chi_H\left(\frac{1}{1-p}\right),$$

where the value at $p = 1$ is defined by polynomial continuation. We classify the inequality $t(H, W) \geq \Phi_H(p)$ on the interval $p \geq 1 - 1/(\chi(H) - 1)$ for every non-bipartite graph H on at most six vertices having neither isolated vertices nor an isolated-edge component. There are 117 such isomorphism classes. The inequality holds for 94 and fails for 23. The positive side combines odd-cycle and pure-chordal theorems, rooted and link inequalities, closure operations, row-specific analytic arguments, and exact rational interval sum-of-squares certificates. Every negative row has an explicit rational step-graphon witness; nineteen come from tensor products of balanced Turán graphons. Lean sources state the row theorem for arbitrary probability spaces and arbitrary measurable symmetric $[0, 1]$ -valued kernels. Exact scripts regenerate the catalogue and the counterexample arithmetic.

1 Introduction

Let $W: \Omega^2 \rightarrow [0, 1]$ be a symmetric measurable kernel on a probability space (Ω, μ) , in the graphon framework of Lovász [6]. For a finite simple graph H , its homomorphism density is

$$t(H, W) = \int_{\Omega^{V(H)}} \prod_{uv \in E(H)} W(x_u, x_v) \prod_{u \in V(H)} d\mu(x_u). \quad (1.1)$$

The edge density is $p = t(K_2, W)$. At a balanced complete k -partite graphon T_k ,

$$p = 1 - \frac{1}{k}, \quad t(H, T_k) = \frac{\chi_H(k)}{k^{v(H)}} = \Phi_H(p). \quad (1.2)$$

This equality motivates the following question.

Definition 1.1. Let H be non-bipartite. We call H *positive* when every graphon W of edge density

$$p \in I_H := \left[1 - \frac{1}{\chi(H) - 1}, 1\right] \quad (1.3)$$

satisfies $t(H, W) \geq \Phi_H(p)$. We call H *negative* when such a graphon and such a p violate the inequality.

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The finite scope considered here consists of the NetworkX Graph Atlas isomorphism classes on at most six vertices that are non-bipartite and have neither isolated vertices nor an isolated-edge component. The exclusions remove factors that do not belong to the connected non-bipartite core of the question. A direct graph-atlas enumeration gives 117 classes.

Theorem 1.2 (Six-vertex classification). *Among the 117 classes in the stated scope, exactly 94 are positive and 23 are negative. The negative Atlas indices are*

$$48, 50, 129, 140, 141, 143, 149, 152, 158, 159, 166, 170, 172, 173, \\ 182, 184, 186, 189, 190, 197, 198, 204, 206. \quad (1.4)$$

Every other class in Table 1 is positive.

The theorem has two independently checkable parts. For a negative row, a finite rational sum evaluates both sides of the proposed inequality. For a positive row, the supplementary Lean theorem quantifies over arbitrary probability spaces and graphons. The proof is therefore not restricted to finite graphs or step graphons.

The reusable mathematics is more informative than the row count. Nineteen positive rows follow from one pure-chordal theorem. Five arise from a theorem for cones over forests, and many others follow from rooted attachment or closure arguments. At the remaining rows, exact certificates replace a large numerical search object by rational identities designed for formal checking. On the negative side, nineteen witnesses have one common tensor construction; only four need local perturbations.

Proof archive. The supplementary archive contains the Lean development, exact certificate files, rational checking scripts, and the generated tables included below. Paths in Table 2 are relative to that archive. The catalogue table is regenerated by `math_paper/scripts/generate_classification.py`; the negative arithmetic is regenerated by `math_paper/scripts/verify_negative_witnesses.py`.

2 The chromatic target

Write $n = v(H)$ and expand the chromatic polynomial as $\chi_H(x) = \sum_{j=0}^n c_j x^j$. Then

$$\Phi_H(p) = \sum_{j=0}^n c_j (1-p)^{n-j}, \quad (2.1)$$

so Φ_H is a polynomial in p and its endpoint value is unambiguous. For a clique, put

$$A_r(p) := \Phi_{K_r}(p) = \prod_{i=1}^{r-1} (ip - (i-1)). \quad (2.2)$$

In particular, $A_2(p) = p$, $A_3(p) = p(2p-1)$, and $A_4(p) = p(2p-1)(3p-2)$.

The target is compatible with several graph operations. If $H = H_1 \sqcup H_2$, then

$$t(H, W) = t(H_1, W)t(H_2, W), \quad \Phi_H(p) = \Phi_{H_1}(p)\Phi_{H_2}(p). \quad (2.3)$$

If H is made from N copies of a graph F by identifying one chosen vertex in every copy, then

$$\chi_H(x) = \frac{\chi_F(x)^N}{x^{N-1}}, \quad \Phi_H(p) = \Phi_F(p)^N. \quad (2.4)$$

If instead the copies share one chosen edge, then

$$\chi_H(x) = \frac{\chi_F(x)^N}{(x(x-1))^{N-1}}, \quad \Phi_H(p) = \frac{\Phi_F(p)^N}{p^{N-1}}. \quad (2.5)$$

These identities follow by fixing the colours on the shared vertex or edge and then colouring the copies independently.

3 Analytic mechanisms for positive rows

This section gives the common inequalities used in the positive classification. The specialized row arguments are indexed in Table 2; their Lean counterparts are identified there as well.

3.1 A weighted rooted triangle inequality

Set

$$d(x) = \int_{\Omega} W(x, y) \, d\mu(y), \quad \tau(x) = \int_{\Omega^2} W(x, y)W(x, z)W(y, z) \, d\mu(y) \, d\mu(z). \quad (3.1)$$

Lemma 3.1 (Weighted rooted Goodman inequality). *If $p > 0$ and h is a nonnegative integer, then*

$$\int_{\Omega} d(x)^h \tau(x) \, d\mu(x) \geq \frac{2p-1}{p} \int_{\Omega} d(x)^{h+2} \, d\mu(x). \quad (3.2)$$

Proof. Put $M_j = \int d(x)^j \, d\mu(x)$, let T_W be the integral operator with kernel W , and define

$$I_h = \int d^h \tau, \quad J_h = \langle d^h, T_W d \rangle.$$

The inequality $ab \geq a + b - 1$ on $[0, 1]^2$, applied to $W(x, y)$ and $W(x, z)$ and then multiplied by $d(x)^h W(y, z)$, gives

$$I_h \geq 2J_h - pM_h. \quad (3.3)$$

Let $U = 1 - W$, $u = T_U \mathbf{1} = 1 - d$, and $q = 1 - p$. Since $T_W d = d - q\mathbf{1} + T_U u$,

$$J_h = M_{h+1} - qM_h + \langle d^h, T_U u \rangle. \quad (3.4)$$

Symmetry of U shows that twice

$$\langle d^h, T_U u \rangle - \int d(x)^h u(x)^2 \, d\mu(x)$$

equals

$$\int_{\Omega^2} U(x, y)(u(y) - u(x))(d(x)^h - d(y)^h) \, d\mu(x) \, d\mu(y).$$

This is nonnegative because $d = 1 - u$. Hence

$$J_h \geq M_{h+2} - M_{h+1} + pM_h.$$

Using this in (3.3) and subtracting the right-hand side of (3.2) gives

$$\frac{1}{p} (M_{h+2} - 2pM_{h+1} + p^2M_h) = \frac{1}{p} \int d(x)^h (d(x) - p)^2 \, d\mu(x) \geq 0.$$

□

This lemma supplies the rooted triangle–tree, book, page, broom, and several leaf-attachment rows in the catalogue. A second local form is also useful. With $B(x) = (T_W d)(x)$,

$$\tau(x) \geq \max\{2B(x) - p, 0\}. \quad (3.5)$$

Indeed, the same inequality $ab \geq a + b - 1$, now integrated only in the two free variables, yields $\tau(x) \geq 2B(x) - p$; nonnegativity supplies the other branch.

3.2 Cones over forests

For $d(x) > 0$, let W_x be W on the probability measure $d\mu_x(y) = W(x, y)d\mu(y)/d(x)$, and let $z_x = t(K_2, W_x) = \tau(x)/d(x)^2$. At $d(x) = 0$ set $z_x = 0$.

Theorem 3.2 (Cones over forests). *Let F be a forest with n vertices, $e \geq 1$ edges, and c connected components. If $H = K_1 \vee F$ and $p \geq 1/2$, then*

$$t(H, W) \geq p^c(2p - 1)^e = \Phi_H(p). \quad (3.6)$$

Proof. The forest case of Sidorenko’s inequality [6, Section 5.2] gives $t(F, W_x) \geq z_x^e$. Conditioning on the image of the universal vertex gives

$$t(H, W) = \int_{\Omega} d(x)^n t(F, W_x) d\mu(x). \quad (3.7)$$

By Lemma 3.1,

$$\frac{\int d(x)^n z_x d\mu(x)}{\int d(x)^n d\mu(x)} \geq 2 - \frac{1}{p}.$$

Apply Jensen’s inequality first to $z \mapsto z^e$ with weight d^n and then to $d \mapsto d^n$. Since a forest has $n = c + e$,

$$t(H, W) \geq p^n \left(2 - \frac{1}{p}\right)^e = p^c(2p - 1)^e.$$

The chromatic-polynomial identity follows by colouring the cone vertex first and then the forest. $\square$

This theorem proves Atlas 40, 111, 117, 135, and 136. Closely related normalized-link arguments prove Atlas 49, 156, and 165. The six cone lifts are Atlas 177, 179, 183, 192, 200, and 205. Atlas 196 follows by applying a normalized-link cone theorem to the exact Atlas 43 inequality.

3.3 Odd cycles and pure chordal graphs

Theorem 3.3 (Odd-cycle bound). *For every odd $m \geq 3$ and every graphon of edge density p ,*

$$t(C_m, W) \geq p^m - p(1 - p)^{m-1} = \Phi_{C_m}(p). \quad (3.8)$$

The proof separates $m = 3, 5, 7$ from $m \geq 9$. For the three small values, Goodman’s triangle inequality [5] is the case $m = 3$. For the three small values, inclusion–exclusion in $U = 1 - W$ reduces the inequality to sums of squares in path moments. For $m \geq 9$, the integral operator of U is split into its constant direction and the mean-zero subspace. The resulting spectral moments are bounded separately on $p \geq 2/3$ and $1/2 < p < 2/3$. This is the proof formalized for Atlas 7 and 38; it is independent of the earlier odd-cycle project. The full derivation is supplied with the archive and is connected to these two row theorems in Table 2.

The second large family is chordal. For $r \geq 3$, a *pure r -chordal graph* is a chordal graph whose maximal cliques all have order r .

Theorem 3.4 (Pure chordal bound). *If H is pure r -chordal, then for every graphon W with $p \geq 1 - 1/(r - 1)$,*

$$t(H, W) \geq \Phi_H(p). \quad (3.9)$$

A clique-tree decomposition writes H as K_r bags joined along clique separators. Entropy gluing bounds the density of the resulting junction by the bag densities divided by the separator densities. The graphon Moon and Moser’s clique-ratio recurrence [7] compares consecutive clique densities, while the chromatic polynomial of the same clique tree factors into the corresponding A_r/A_s terms from (2.2); see also Agnarsson’s factorization for chordal graphs [1]. Multiplying these comparisons proves (3.9). A regularization step removes divisions by zero. This proof gives the nineteen pure-chordal rows listed in Table 1 and does not require connectedness.

3.4 Closure and row-specific arguments

Three closure rules reduce further rows to smaller ones.

Proposition 3.5 (Closure rules). *Suppose all target functions below are nonnegative on the interval under consideration.*

- (i) *Positivity is preserved under disjoint union.*
- (ii) *If H is a vertex self-amalgam of N copies of F , then $t(H, W) \geq t(F, W)^N$, and (2.4) transfers the bound from F to H .*
- (iii) *If H is an edge self-amalgam of N copies of F , then a rooted Cauchy–Schwarz argument gives $t(H, W) \geq t(F, W)^N/p^{N-1}$, matching (2.5).*
- (iv) *Let $\text{Wh}(F)$ attach one leaf to every vertex of F . If Φ_F is nondecreasing on the required interval, then*

$$t(\text{Wh}(F), W) \geq p^{v(F)} \Phi_F(p) = \Phi_{\text{Wh}(F)}(p).$$

The proof of each item conditions on the common rooted variables and applies Hölder, Cauchy–Schwarz, or Jensen as appropriate. In the whiskering case, conditioning on the original vertices inserts one degree factor per vertex; a change of measure followed by monotonicity of Φ_F yields the stated bound.

The remaining analytic rows use a short list of two-dimensional reductions: supporting planes for rooted density profiles, an exact square for Atlas 145, a Hilbert-space projection for Atlas 148, and a concrete clique–odd-cycle inequality for Atlas 187. The Atlas 126 and 148 arguments use the triangle-density profile proved by Fisher [4] and subsumed by Reiher’s clique-density theorem [8]. The triangle three-edge-tail argument uses the odd-walk inequality of Blekherman and Raymond [3]. Each reduction ends in a univariate polynomial inequality on the required interval. The polynomial steps are checked with rational factorization or Bernstein coefficients. The row-to-proof index in Table 2 distinguishes these methods from the exact rooted certificates in the next section.

4 Exact rooted certificates

The difficult positive rows are proved by rational identities over rooted graph densities. We briefly describe the common semantics.

Choose labelled root vertices and sample their images in Ω . Each possible edge is represented by a Bernoulli variable whose conditional mean is the corresponding value of W . Repeated occurrences of the same labelled edge use the same Bernoulli variable. This convention is essential: using independent copies would replace a factor W by W^2 and would not represent the graph density.

For a finite vector v of rooted expressions and a rational positive semidefinite matrix Q , the conditional expectation of $v^\top Q v$ is nonnegative. An interval certificate combines such terms with the nonnegative factors $p - a$ and $b - p$:

$$t(H, W) - \Phi_H(p) = \mathbb{E} \left[v_0^\top Q_0 v_0 + (p - a) v_1^\top Q_1 v_1 + (b - p) v_2^\top Q_2 v_2 + R \right], \quad (4.1)$$

where R is a rational linear combination already known to be nonnegative. Every equality in (4.1) is checked coefficientwise, and each Q_i is supplied by a rational Gram factorization or a strict diagonal dominance proof.

Proposition 4.1 (Certificate rows). *The following exact certificates prove the indicated rows throughout their required intervals.*

- (i) *Atlas 43: a three-root identity on $[1/2, 1]$ with 91 nonzero coefficients and two rational Gram corrections of orders 107 and 48.*
- (ii) *Atlas 118, 122, 124, 147, 151, 153, 157, 168, 169, 171, 174, 181, 185, 188, 194, 199: four-root interval identities, each with 407 coefficient equations.*
- (iii) *Atlas 130 on $[1/2, 2/3]$ and Atlas 203 on $[2/3, 3/4]$: induced rooted identities using 28 positive integer Gram factors and 936 coefficient equations. Four-root identities cover $[2/3, 1]$ for Atlas 130 and $[3/4, 1]$ for Atlas 203.*
- (iv) *Atlas 127: a direct ground-6 split certificate on $[1/2, 1]$.*

The direct ground-6 certificate in part (iv) is the proof used for Atlas 127 in this classification. A stronger smoothed-Goodman statement was also investigated, but it is not needed for the row theorem and is not used here. Atlas 43 enters Atlas 196 through the normalized-link implication mentioned after Theorem 3.2.

The numerical searches that suggested these identities are not proof inputs. The final files contain rational coefficients, exact polynomial equalities, and nonnegativity decompositions. The Lean development connects their rooted semantics to (1.1).

5 Exact counterexamples

Let $\mu = (\mu_1, \dots, \mu_s)$ be rational nonnegative masses summing to one, and let $A \in [0, 1]^{s \times s}$ be a symmetric rational matrix. The associated step graphon satisfies

$$p = \sum_{i,j=1}^s \mu_i \mu_j A_{ij}, \quad (5.1)$$

$$t(H, W) = \sum_{f: V(H) \rightarrow [s]} \prod_{u \in V(H)} \mu_{f(u)} \prod_{uv \in E(H)} A_{f(u), f(v)}. \quad (5.2)$$

Thus every counterexample in Table 3 is a finite exact rational computation.

5.1 Tensor products of Turán graphons

On $[a] \times [b]$ with the uniform measure, define

$$(T_a \otimes T_b)((i, j), (i', j')) = \mathbf{1}_{i \neq i'} \mathbf{1}_{j \neq j'}. \quad (5.3)$$

Lemma 5.1 (Tensor evaluation). *For every finite graph H ,*

$$p = \left(1 - \frac{1}{a}\right) \left(1 - \frac{1}{b}\right), \quad t(H, T_a \otimes T_b) = \frac{\chi_H(a) \chi_H(b)}{(ab)^{v(H)}}. \quad (5.4)$$

Proof. An edge is present precisely when its two coordinates differ in both positions, which gives the formula for p . A homomorphism to the step graph is exactly an ordered pair of proper colourings, one with a colours and one with b colours. There are $\chi_H(a) \chi_H(b)$ such pairs among $(ab)^{v(H)}$ maps. $\square$

Substitution into (5.4) proves strict failure for nineteen of the rows in (1.4). The exact values appear in Table 3.

5.2 Four local witnesses

Atlas 166 and 172 use uniform masses on four parts and

$$A = \frac{1}{4} \begin{pmatrix} 0 & 1 & 4 & 4 \\ 1 & 0 & 4 & 4 \\ 4 & 4 & 0 & 1 \\ 4 & 4 & 1 & 0 \end{pmatrix}. \quad (5.5)$$

Atlas 206 uses uniform masses on three parts and

$$A = \frac{1}{4} \begin{pmatrix} 1 & 4 & 4 \\ 4 & 0 & 4 \\ 4 & 4 & 0 \end{pmatrix}. \quad (5.6)$$

These give $p = 9/16$ and $p = 25/36$, respectively.

For Atlas 152, use the five masses

$$\mu = \frac{1}{3750000} (1870752, 2998, 1875000, 125, 1125) \quad (5.7)$$

and

$$A = \frac{1}{16} \begin{pmatrix} 0 & 0 & 16 & 16 & 0 \\ 0 & 0 & 16 & 0 & 16 \\ 16 & 16 & 0 & 1 & 16 \\ 16 & 0 & 1 & 0 & 0 \\ 0 & 16 & 16 & 0 & 0 \end{pmatrix}. \quad (5.8)$$

Equations (5.1)–(5.2) give the strict rational gaps in Table 3. This finishes the negative half of Theorem 1.2.

6 Formal verification

The Lean definition of a graphon uses an arbitrary probability space and a symmetric measurable kernel with values in $[0, 1]$. For each Atlas row, the example module proves either `SatisfiesLowerBound` or `ViolatesLowerBound`. A source scan finds all 117 modules marked `.verified` and no occurrence of `sorry` in those modules.

The formal proof is intentionally row-oriented. Some analytic notes state a more general theorem, while Lean proves only the specialization needed for a catalogue row. The pure-chordal, rooted-SOS, and common negative constructions have reusable method-level formalizations; several specialized analytic rows instead expose only their final row theorem. Table 2 names the mathematical and Lean source for every row, so these scopes can be checked without inferring a general theorem from a row specialization.

Twenty compact or specialized rows have retained sequential-build measurements, reproduced in Table 4. Their combined time was 701.5 minutes and the largest measured peak memory was 11.91 GiB. Atlas 43 used a separate 1097-module sequential check: 6.02 hours and 15.4 GiB peak memory. A parallel attempt on a 64 GiB machine exhausted memory. These measurements concern proof checking, not the mathematical truth of the identities.

The author reports a successful full verification of the complete development on another device. The directory from that run and its exact start and end times are unavailable in the present archive. We therefore separate that author-attested full build from the retained row-level measurements and from the reproducible source scan.

7 Proof of the classification theorem

Proof of Theorem 1.2. The generation script enumerates the Graph Atlas, applies the stated scope conditions, and independently checks each graph6 identifier, vertex count, edge count, chromatic number, and chromatic polynomial. It obtains 117 rows.

For the 23 indices in (1.4), the rational checker uses Lemma 5.1 or (5.1)–(5.8) and verifies $t(H, W) - \Phi_H(p) < 0$. All witness densities lie in I_H .

For each remaining row, Table 1 assigns one of the positive methods in Sections 3–4, and Table 2 identifies the corresponding mathematical source and Lean theorem. The generated audit checks that each of these 94 modules has theorem kind `SatisfiesLowerBound`; the formal theorem unfolds to the inequality in Definition 1.1 for every graphon and every $p \in I_H$. The two sets are disjoint and contain all 117 rows. Hence 94 rows are positive and 23 are negative. $\square$

8 Discussion

The classification has three mathematical features. First, most positive rows do not need case-specific semidefinite searches: clique-tree gluing, rooted Goodman inequalities, links, and closure operations cover broad families. Second, the hard positive cases admit compact exact replacements for much larger search outputs. Third, the negative side is even more uniform: tensor products of Turán graphons settle nineteen of twenty-three rows.

The order-six boundary should not be interpreted as a claim about all finite graphs. Nor does the present work supply a simple graph-theoretic characterization of positivity. Its conclusion is a complete finite classification together with reusable sufficient conditions and explicit obstructions. The machine-checked artifacts also make clear which claims are general graphon theorems and which are formalized only at a fixed Atlas row.

AI assistance disclosure. OpenAI Codex drafted this manuscript from the project sources and generated some of the organizational prose. The mathematical claims are tied to the exact scripts, certificates, and Lean sources described above. The authors remain responsible for the submitted text and conclusions. A separate companion article studies the use of ChatGPT Pro, Codex, and Claude Code in the proof campaign [2].

A Complete classification

Table 1: Complete classification. The interval column gives the required range of edge densities.

Atlas graph6 v	e	χ	interval	$\Phi_H(p)$	method	Lean theorem
7 Bw	3	3	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)$	+ odd-cycle bound	Graph007.status
15 CN	4	4	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1)$	+ paw inequality	Graph015.status
17 Cl	4	5	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)^2$	+ pure chordal	Graph017.status
18 C-	4	6	$4 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)(3p-2)$	+ pure chordal	Graph018.status
34 D0f	5	5	$3 \left[\frac{1}{2}, 1 \right]$	$p^3 \cdot (2p-1)$	+ rooted triangle-tree inequality	Graph034.status
35 Dx	5	5	$3 \left[\frac{1}{2}, 1 \right]$	$p^3 \cdot (2p-1)$	+ two-root book inequality	Graph035.status
36 DJc	5	5	$3 \left[\frac{1}{2}, 1 \right]$	$p^3 \cdot (2p-1)$	+ rooted triangle-tree inequality	Graph036.status
38 Dhc	5	5	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1) \left(2p^2 - 2p + 1 \right)$	+ odd-cycle bound	Graph038.status
40 DjW	5	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^2(2p-1)^2$	+ cone over a forest	Graph040.status
41 Db	5	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^2(2p-1)^2$	+ rooted triangle-tree inequality	Graph041.status
42 D'f	5	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^2(2p-1)^2$	+ pure chordal	Graph042.status
43 Dlc	5	6	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1) \left(3p^2 - 3p + 1 \right)$	+ three-root interval SOS	Graph043.status
45 DJf	5	7	$4 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1)(3p-2)$	+ clique common-leaf inequality	Graph045.status
46 DFf	5	7	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)^3$	+ pure chordal	Graph046.status
47 Djs	5	7	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)^3$	+ pure chordal	Graph047.status
48 DJw	5	7	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1) \left(5p^2 - 6p + 2 \right)$	- tensor product of Turan graphons	Graph048.status
49 Df	5	8	$4 \left[\frac{2}{3}, 1 \right]$	$p(2p-1)^2 \cdot (3p-2)$	+ special cone inequality	Graph049.status
50 Dlf	5	8	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1) \left(7p^2 - 9p + 3 \right)$	- tensor product of Turan graphons	Graph050.status
51 Dnf	5	9	$4 \left[\frac{2}{3}, 1 \right]$	$p(2p-1)(3p-2)^2$	+ pure chordal	Graph051.status
52 D-f	5	10	$5 \left[\frac{2}{3}, 1 \right]$	$p(2p-1)(3p-2)(4p-3)$	+ pure chordal	Graph052.status
84 ECd	6	5	$3 \left[\frac{1}{2}, 1 \right]$	$p^3 \cdot (2p-1)$	+ component multiplicativity	Graph084.status
92 E7Fv	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ rooted triangle-tree inequality	Graph092.status
93 ECf0	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ two-root book inequality	Graph093.status
94 E0dW	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ whiskering	Graph094.status
95 EGf?	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ mixed rooted-triangle inequality	Graph095.status
97 EYWO	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ adjacent leaf-tail inequality	Graph097.status
100 EsCW	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ triangle broom inequality	Graph100.status
102 EJe?	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^4 \cdot (2p-1)$	+ triangle tail inequality	Graph102.status
104 Ehd?	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1) \left(2p^2 - 2p + 1 \right)$	+ odd-cycle leaf inequality	Graph104.status
106 EJaG	6	6	$3 \left[\frac{1}{2}, 1 \right]$	$p^2(2p-1)^2$	+ component multiplicativity	Graph106.status
111 EBfG	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ cone over a forest	Graph111.status
112 EhX	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ two-root book inequality	Graph112.status
113 Ef-O	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ diamond leaf inequality	Graph113.status
114 EJwG	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ rooted triangle-tree inequality	Graph114.status
115 E'Xg	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ self-amalgamation	Graph115.status
117 EtaG	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ cone over a forest	Graph117.status
118 Eht?	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1) \left(3p^2 - 3p + 1 \right)$	+ Compact four-root SOS	Graph118.status
119 Eto0	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ bowtie leaf inequality	Graph119.status
120 EBf?	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ triangle-book attachment in-equality	Graph120.status
122 Eld?	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1) \left(3p^2 - 3p + 1 \right)$	+ Compact four-root SOS	Graph122.status
123 EJy?	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^3(2p-1)^2$	+ triangle-book attachment in-equality	Graph123.status
124 Exd?	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1) \left(3p^2 - 3p + 1 \right)$	+ Compact four-root SOS	Graph124.status
126 ERUO	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p^2 \cdot (2p-1) \left(3p^2 - 3p + 1 \right)$	+ Atlas 126 triangle-SC_4\$ vertex supporting-plane theorem	Graph126.status
127 EZEG	6	7	$3 \left[\frac{1}{2}, 1 \right]$	$p(2p-1)^2 \cdot \left(2p^2 - 2p + 1 \right)$	+ direct compact interval SOS	Graph127.status

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Table 1 (continued)

Atlas graph6	v	e	χ interval	$\Phi_H(p)$	method	Lean theorem
129 Ehe0	6	7	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1) \left(5p^3 - 8p^2 + 5p - 1\right)$	– tensor product of Turan graphons Graph129.status
130 E{CW	6	7	3	$\left[\frac{1}{2}, 1\right]$	$p^3(2p-1)^2$	+ Induced and four-root SOS Graph130.status
133 E-a?	6	8	4	$\left[\frac{2}{3}, 1\right]$	$p^3 \cdot (2p-1)(3p-2)$	+ clique common-leaf inequality Graph133.status
134 E-_0	6	8	4	$\left[\frac{2}{3}, 1\right]$	$p^3 \cdot (2p-1)(3p-2)$	+ clique distributed-leaf inequality Graph134.status
135 EzW_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ cone over a forest Graph135.status
136 Ejt?	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ cone over a forest Graph136.status
137 EjsG	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ triangle-book attachment in-equality Graph137.status
138 Ez'_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ rooted triangle-tree inequality Graph138.status
139 Eju?	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ triangle-book attachment in-equality Graph139.status
140 Ev'_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2 \cdot (2p-1) \left(5p^2 - 6p + 2\right)$	– tensor product of Turan graphons Graph140.status
141 EXSw	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2 \cdot (2p-1) \left(5p^2 - 6p + 2\right)$	– tensor product of Turan graphons Graph141.status
142 E-AG	6	8	4	$\left[\frac{2}{3}, 1\right]$	$p^3 \cdot (2p-1)(3p-2)$	+ supporting-plane inequality Graph142.status
143 Er'°	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2 \cdot (2p-1) \left(5p^2 - 6p + 2\right)$	– tensor product of Turan graphons Graph143.status
144 EB}G	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ pure chordal Graph144.status
145 Exe_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(3p^2 - 3p + 1\right)$	+ exact square identity Graph145.status
147 EhMg	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(3p^2 - 3p + 1\right)$	+ Compact four-root SOS Graph147.status
148 EyUG	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(3p^2 - 3p + 1\right)$	+ Hilbert projection Graph148.status
149 Ele_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1) \left(7p^3 - 11p^2 + 6p - 1\right)$	– tensor product of Turan graphons Graph149.status
150 EJyG	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p^2(2p-1)^3$	+ pure chordal Graph150.status
151 EhdW	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(4p^2 - 5p + 2\right)$	+ Compact four-root SOS Graph151.status
152 EhNG	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(3p^2 - 3p + 1\right)$	– fractional-fibre step graphon Graph152.status
153 Ehf_	6	8	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1) \left(7p^3 - 12p^2 + 8p - 2\right)$	+ Compact four-root SOS Graph153.status
156 E-H_	6	9	4	$\left[\frac{2}{3}, 1\right]$	$p^2(2p-1)^2 \cdot (3p-2)$	+ special cone inequality Graph156.status
157 E-'_	6	9	4	$\left[\frac{2}{3}, 1\right]$	$p^2(2p-1)^2 \cdot (3p-2)$	+ Compact four-root SOS Graph157.status
158 El{G	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p^2 \cdot (2p-1) \left(7p^2 - 9p + 3\right)$	– tensor product of Turan graphons Graph158.status
159 EZSw	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p^2 \cdot (2p-1) \left(7p^2 - 9p + 3\right)$	– tensor product of Turan graphons Graph159.status
160 E-@g	6	9	4	$\left[\frac{2}{3}, 1\right]$	$p^2(2p-1)^2 \cdot (3p-2)$	+ supporting-plane inequality Graph160.status
161 E?-w	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^4$	+ pure chordal Graph161.status
162 El e_	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^4$	+ pure chordal Graph162.status
163 EyuG	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^4$	+ pure chordal Graph163.status
164 EyVG	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^4$	+ pure chordal Graph164.status
165 E-aG	6	9	4	$\left[\frac{2}{3}, 1\right]$	$p^2(2p-1)^2 \cdot (3p-2)$	+ special cone inequality Graph165.status
166 Elf0	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(5p^2 - 6p + 2\right)$	– local perturbation of a Turan graphon Graph166.status
167 E~eG	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^4$	+ pure chordal Graph167.status
168 E~MG	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1)^2 \cdot \left(5p^2 - 6p + 2\right)$	+ Compact four-root SOS Graph168.status
169 Exf_	6	9	4	$\left[\frac{2}{3}, 1\right]$	$p(2p-1)(3p-2) \left(3p^2 - 3p + 1\right)$	+ Compact four-root SOS Graph169.status
170 EO-°	6	9	3	$\left[\frac{1}{2}, 1\right]$	$p(2p-1) \left(11p^3 - 19p^2 + 11p - 2\right)$	– tensor product of Turan graphons Graph170.status

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Table 1 (continued)

Atlas graph6	v	e	χ interval	$\Phi_H(p)$	method	Lean theorem
171 Ehew	6	9	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(11p^3 - 20p^2 + 13p - 3 \right)$	+ Compact four-root SOS	Graph171.status
172 Elf ₋	6	9	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(5p^2 - 6p + 2 \right)$	- local perturbation of a Turan graphon	Graph172.status
173 ElMg	6	9	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(6p^2 - 8p + 3 \right)$	- tensor product of Turan graphons	Graph173.status
174 EtTg	6	9	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(13p^3 - 25p^2 + 17p - 4 \right)$	+ Compact four-root SOS	Graph174.status
177 EnG	6	10	4 $[\frac{2}{3}, 1]$	$p^2 \cdot (2p-1) (3p-2)^2$	+ six explicit cone lifts	Graph177.status
178 En}?	6	10	4 $[\frac{2}{3}, 1]$	$p^2 \cdot (2p-1) (3p-2)^2$	+ supporting-plane inequality	Graph178.status
179 E ₋ w	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1)^3 \cdot (3p-2)$	+ six explicit cone lifts	Graph179.status
180 EjtW	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1)^3 \cdot (3p-2)$	+ two-root book inequality	Graph180.status
181 E ⁻ mG	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1)^3 \cdot (3p-2)$	+ Compact four-root SOS	Graph181.status
182 E ⁻ Mg	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(7p^2 - 9p + 3 \right)$	- tensor product of Turan graphons	Graph182.status
183 EjvG	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1)^3 \cdot (3p-2)$	+ six explicit cone lifts	Graph183.status
184 Elfo	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(7p^2 - 9p + 3 \right)$	- tensor product of Turan graphons	Graph184.status
185 Exv ₋	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(5p^2 - 6p + 2 \right)$	+ Compact four-root SOS	Graph185.status
186 ErXw	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(8p^2 - 11p + 4 \right)$	- tensor product of Turan graphons	Graph186.status
187 Ehf _w	6	10	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(5p^2 - 6p + 2 \right)$	+ concrete clique-odd-cycle theorem	Graph187.status
188 EzNG	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(17p^3 - 33p^2 + 22p - 5 \right)$	+ Compact four-root SOS	Graph188.status
189 E ⁻ NG	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(19p^3 - 38p^2 + 26p - 6 \right)$	- tensor product of Turan graphons	Graph189.status
190 EyUw	6	10	3 $[\frac{1}{2}, 1]$	$p(2p-1)^2 \cdot \left(8p^2 - 11p + 4 \right)$	- tensor product of Turan graphons	Graph190.status
191 E- ?	6	11	5 $[\frac{3}{4}, 1]$	$p^2 \cdot (2p-1) (3p-2) (4p-3)$	+ clique common-leaf inequality	Graph191.status
192 E-Xo	6	11	4 $[\frac{3}{4}, 1]$	$p(2p-1)^2 (3p-2)^2$	+ six explicit cone lifts	Graph192.status
193 En}G	6	11	4 $[\frac{3}{4}, 1]$	$p(2p-1)^2 (3p-2)^2$	+ rooted triangle-tree inequality	Graph193.status
194 E-wW	6	11	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(7p^2 - 9p + 3 \right)$	+ Compact four-root SOS	Graph194.status
195 EyVw	6	11	4 $[\frac{2}{3}, 1]$	$p(2p-1)^2 (3p-2)^2$	+ pure chordal	Graph195.status
196 ER-g	6	11	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(7p^2 - 9p + 3 \right)$	+ normalized-link cone lift	Graph196.status
197 E}^G	6	11	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(23p^3 - 46p^2 + 31p - 7 \right)$	- tensor product of Turan graphons	Graph197.status
198 Ep-o	6	11	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(23p^3 - 46p^2 + 31p - 7 \right)$	- tensor product of Turan graphons	Graph198.status
199 El ⁻ g	6	11	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(8p^2 - 11p + 4 \right)$	+ Compact four-root SOS	Graph199.status
200 E-{W	6	12	5 $[\frac{3}{4}, 1]$	$p(2p-1)^2 \cdot (3p-2) (4p-3)$	+ six explicit cone lifts	Graph200.status
201 E-z ₋	6	12	4 $[\frac{3}{4}, 1]$	$p(2p-1) (3p-2)^3$	+ pure chordal	Graph201.status
202 Ep-w	6	12	4 $[\frac{3}{4}, 1]$	$p(2p-1) (3p-2)^3$	+ pure chordal	Graph202.status
203 E ⁻ G	6	12	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(10p^2 - 14p + 5 \right)$	+ Induced and four-root SOS	Graph203.status
204 EznW	6	12	3 $[\frac{1}{2}, 1]$	$p(2p-1) \left(32p^3 - 67p^2 + 47p - 11 \right)$	- tensor product of Turan graphons	Graph204.status
205 E--G	6	13	5 $[\frac{3}{4}, 1]$	$p(2p-1) (3p-2)^2 \cdot (4p-3)$	+ six explicit cone lifts	Graph205.status
206 E-nW	6	13	4 $[\frac{2}{3}, 1]$	$p(2p-1) (3p-2) \left(13p^2 - 19p + 7 \right)$	- local perturbation of a Turan graphon	Graph206.status
207 Ez-w	6	14	5 $[\frac{3}{4}, 1]$	$p(2p-1) (3p-2) (4p-3)^2$	+ pure chordal	Graph207.status

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Table 1 (continued)						
Atlas graph6	v	e	χ interval	$\Phi_H(p)$	method	Lean theorem
208 E--w	6	15	$6 \left[\frac{4}{5}, 1\right]$	$p(2p-1)(3p-2)(4p-3)(5p-4)$	+ pure chordal	Graph208.status

B Row-to-source index

Table 2: Machine-readable identifiers, edge lists, and source files.

Atlas	graph6	edge list	source
7	Bw	01, 02, 12	notes/paper_new_region2_v3.tex; lean/Taeyoung/Examples/Graph007.lean
15	CN	03, 12, 13, 23	notes/rooted_triangle_tree_extensions.tex; lean/Taeyoung/Examples/Graph015.lean
17	C	01, 02, 03, 12, 23	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph017.lean
18	C~	01, 02, 03, 12, 13, 23	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph018.lean
34	D@{	04, 14, 23, 24, 34	notes/rooted_triangle_tree_extensions.tex; lean/Taeyoung/Examples/Graph034.lean
35	Dx_	01, 02, 04, 12, 23	notes/two_root_triangle_leaf_cones.tex; lean/Taeyoung/Examples/Graph035.lean
36	DJc	04, 12, 13, 23, 34	notes/rooted_triangle_tree_extensions.tex; lean/Taeyoung/Examples/Graph036.lean
38	Dhc	01, 04, 12, 23, 34	notes/paper_new_region2_v3.tex; lean/Taeyoung/Examples/Graph038.lean
40	DjW	01, 12, 13, 14, 23, 24	notes/forest_cone_graphon_bound.tex; lean/Taeyoung/Examples/Graph040.lean
41	Db[	01, 13, 14, 23, 24, 34	notes/page_rooted_triangle_book_leaves.tex; lean/Taeyoung/Examples/Graph041.lean
42	D'{'	01, 04, 14, 23, 24, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph042.lean
43	Dlc	01, 03, 04, 12, 23, 34	notes/atlas43_exact_rooted_sos.tex; lean/Taeyoung/Examples/Graph043.lean
45	DJ{	04, 12, 13, 14, 23, 24, 34	notes/cliue_common_leaf_extensions.tex; lean/Taeyoung/Examples/Graph045.lean
46	DF{	03, 04, 13, 14, 23, 24, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph046.lean
47	Djs	01, 04, 12, 13, 14, 23, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph047.lean
48	D]w	02, 03, 04, 12, 13, 14, 24	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/verify_negative_witnesses.py; lean/Taeyoung/Examples/Graph048.lean
49	Df{	01, 03, 04, 13, 14, 23, 24, 34	notes/paw_triangle_edge_cones.tex; lean/Taeyoung/Examples/Graph049.lean
50	Dl{	01, 03, 04, 12, 14, 23, 24, 34	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/verify_negative_witnesses.py; lean/Taeyoung/Examples/Graph050.lean

Atlas	graph6	edge list	source
51	Dn{	01, 03, 04, 12, 13, 14, 23, 24, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph051.lean
52	D~{	01, 02, 03, 04, 12, 13, 14, 23, 24, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph052.lean
84	ECd_	03, 04, 15, 25, 34	history/paper_independent/math_paper/main.tex; lean/Taeyoung/Examples/Graph084.lean
92	E?Fw	05, 15, 25, 34, 35, 45	notes/rooted_triangle_tree_extensions.tex; lean/Taeyoung/Examples/Graph092.lean
93	EC{0	03, 04, 14, 24, 34, 35	notes/two_root_triangle_leaf_cones.tex; lean/Taeyoung/Examples/Graph093.lean
94	E@dW	04, 15, 23, 34, 35, 45	notes/whiskering.tex; lean/Taeyoung/Examples/Graph094.lean
95	EG}??	04, 05, 12, 14, 24, 34	notes/mixed_rooted_triangle_branches.tex; lean/Taeyoung/Examples/Graph095.lean
97	EYWO	02, 12, 13, 14, 24, 35	notes/triangle_adjacent_leaf_tail.tex; lean/Taeyoung/Examples/Graph097.lean
100	EsCW	01, 02, 03, 34, 35, 45	notes/triangle_two_leaf_broom.tex; lean/Taeyoung/Examples/Graph100.lean
102	EJe?	04, 05, 12, 13, 23, 34	notes/triangle_three_edge_tail.tex; lean/Taeyoung/Examples/Graph102.lean
104	Ehd?	01, 04, 12, 15, 23, 34	notes/odd_cycle_one_leaf.tex; lean/Taeyoung/Examples/Graph104.lean
106	EJaG	04, 05, 12, 13, 23, 45	history/paper_independent/math_paper/main.tex; lean/Taeyoung/Examples/Graph106.lean
111	EB{G	04, 13, 14, 23, 24, 34, 45	notes/forest_cone_graphon_bound.tex; lean/Taeyoung/Examples/Graph111.lean
112	EhX_	01, 12, 14, 15, 23, 24, 25	notes/two_root_triangle_leaf_cones.tex; lean/Taeyoung/Examples/Graph112.lean
113	E^_0	02, 03, 04, 12, 13, 23, 35	notes/diamond_orbit_balanced_leaves.tex; lean/Taeyoung/Examples/Graph113.lean
114	EJwG	04, 12, 13, 14, 23, 24, 45	notes/page_rooted_triangle_book_leaves.tex; lean/Taeyoung/Examples/Graph114.lean
115	E'Xg	01, 14, 15, 23, 24, 25, 45	notes/self_amalgam.tex; lean/Taeyoung/Examples/Graph115.lean
117	EtaG	01, 02, 03, 04, 05, 23, 45	notes/forest_cone_graphon_bound.tex; lean/Taeyoung/Examples/Graph117.lean
118	Eht?	01, 04, 12, 14, 15, 23, 34	notes/atlas118_compact_certificate.md; lean/Taeyoung/Examples/Graph118.lean
119	Eto0	01, 02, 03, 04, 14, 23, 35	notes/bowtie_outer_leaves.tex; lean/Taeyoung/Examples/Graph119.lean
120	EB}??	04, 05, 13, 14, 23, 24, 34	notes/triangle_book_two_edge_tail.tex; lean/Taeyoung/Examples/Graph120.lean

Atlas	graph6	edge list	source
122	Eld?	01, 03, 04, 12, 15, 23, 34	notes/atlas122_compact_certificate.md; lean/Taeyoung/Examples/Graph122.lean
123	EJy?	04, 05, 12, 13, 14, 23, 24	notes/triangle_book_page_two_edge_tail.tex; lean/Taeyoung/Examples/Graph123.lean
124	Exd?	01, 02, 04, 12, 15, 23, 34	notes/atlas124_compact_certificate.md; lean/Taeyoung/Examples/Graph124.lean
126	ERUO	02, 05, 13, 14, 23, 34, 35	notes/atlas126_compact_verification.md; notes/atlas126_triangle_c4_vertex_supporting_plane.tex; lean/Taeyoung/Examples/Graph126.lean
127	EZEG	02, 05, 12, 13, 23, 34, 45	notes/atlas127_compact_certificate.md; notes/smoothed_goodman_deflagged.tex; lean/Taeyoung/Examples/Graph127.lean
129	EheO	01, 04, 05, 12, 23, 34, 35	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph129.lean
130	E{CW	01, 02, 03, 12, 34, 35, 45	notes/atlas130_compact_certificate.md; lean/Taeyoung/Examples/Graph130.lean
133	E~a?	01, 02, 03, 04, 05, 12, 13, 23	notes/clique_common_leaf_extensions.tex; lean/ Taeyoung/Examples/Graph133.lean
134	E~_O	01, 02, 03, 04, 12, 13, 23, 35	notes/clique_distributed_leaves.tex; lean/ Taeyoung/Examples/Graph134.lean
135	EzW_	01, 02, 12, 13, 14, 23, 24, 25	notes/forest_cone_graphon_bound.tex; lean/ Taeyoung/Examples/Graph135.lean
136	Ejt?	01, 04, 12, 13, 14, 15, 23, 34	notes/forest_cone_graphon_bound.tex; lean/ Taeyoung/Examples/Graph136.lean
137	EjsG	01, 04, 12, 13, 14, 23, 34, 45	notes/triangle_book_page_paw_branch.tex; lean/ Taeyoung/Examples/Graph137.lean
138	Ez' _	01, 02, 04, 12, 13, 15, 23, 25	notes/page_rooted_triangle_book_leaves.tex; lean/Taeyoung/Examples/Graph138.lean
139	Eju?	01, 04, 05, 12, 13, 14, 23, 34	notes/triangle_book_page_paw_branch.tex; lean/ Taeyoung/Examples/Graph139.lean
140	Ev' _	01, 02, 03, 04, 13, 15, 23, 25	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph140.lean
141	EXSw	02, 12, 14, 23, 25, 34, 35, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph141.lean
142	E~AG	01, 02, 03, 05, 12, 13, 23, 45	notes/k4_two_edge_tail.tex; lean/Taeyoung/ Examples/Graph142.lean

Atlas	graph6	edge list	source
143	Er' o	01, 02, 04, 13, 15, 23, 25, 35	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph143.lean
144	EB}G	04, 05, 13, 14, 23, 24, 34, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph144.lean
145	Exe_	01, 02, 04, 05, 12, 23, 25, 34	notes/c4_triangle_page_concentration.tex; lean/ Taeyoung/Examples/Graph145.lean
147	EhMg	01, 05, 12, 23, 24, 25, 34, 45	notes/atlas147_compact_certificate.md; lean/Taeyoung/Examples/Graph147.lean
148	EyUG	01, 02, 05, 12, 13, 14, 34, 45	notes/atlas148_paw_bias_hilbert_projection.tex; lean/Taeyoung/Examples/Graph148.lean
149	Ele_	01, 03, 04, 05, 12, 23, 25, 34	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph149.lean
150	EJyG	04, 05, 12, 13, 14, 23, 24, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph150.lean
151	EhdW	01, 04, 12, 15, 23, 34, 35, 45	notes/atlas151_compact_certificate.md; lean/Taeyoung/Examples/Graph151.lean
152	EhNG	01, 05, 12, 15, 23, 24, 34, 45	notes/atlas152_fractional_fibre_local_ counterexample.tex; lean/Taeyoung/Examples/ Graph152.lean
153	Ehf_	01, 04, 05, 12, 15, 23, 25, 34	notes/atlas153_compact_certificate.md; lean/Taeyoung/Examples/Graph153.lean
156	E~H_	01, 02, 03, 12, 13, 15, 23, 24, 25	notes/paw_triangle_edge_cones.tex; lean/ Taeyoung/Examples/Graph156.lean
157	E~' _	01, 02, 03, 04, 12, 13, 15, 23, 25	notes/atlas157_compact_certificate.md; lean/Taeyoung/Examples/Graph157.lean
158	El{G	01, 03, 04, 12, 14, 23, 24, 34, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph158.lean
159	EZSw	02, 12, 13, 14, 23, 25, 34, 35, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph159.lean
160	E~@g	01, 02, 03, 12, 13, 15, 23, 25, 45	notes/atlas160_k4_paw_edge_supporting_plane. tex; lean/Taeyoung/Examples/Graph160.lean
161	E?~w	04, 05, 14, 15, 24, 25, 34, 35, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph161.lean
162	E e_	01, 02, 03, 04, 05, 12, 23, 25, 34	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph162.lean

Atlas	graph6	edge list	source
163	EyuG	01, 02, 04, 05, 12, 13, 14, 34, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph163.lean
164	EyVG	01, 02, 05, 12, 13, 14, 15, 34, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph164.lean
165	E~aG	01, 02, 03, 04, 05, 12, 13, 23, 45	notes/paw_triangle_edge_cones.tex; lean/Taeyoung/Examples/Graph165.lean
166	Elf0	01, 03, 04, 05, 12, 15, 23, 34, 35	notes/turan_local_and_high_density_negative_tests.tex; lean/Taeyoung/Examples/Graph166.lean
167	E^eG	02, 03, 04, 05, 12, 13, 23, 34, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/Graph167.lean
168	E^MG	02, 03, 05, 12, 13, 23, 24, 34, 45	notes/atlas168_compact_certificate.md; lean/Taeyoung/Examples/Graph168.lean
169	Exf_	01, 02, 04, 05, 12, 15, 23, 25, 34	notes/atlas169_compact_certificate.md; lean/Taeyoung/Examples/Graph169.lean
170	EO~o	02, 04, 05, 14, 15, 24, 25, 34, 35	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/verify_negative_witnesses.py; lean/Taeyoung/Examples/Graph170.lean
171	Ehew	01, 04, 05, 12, 23, 25, 34, 35, 45	notes/atlas171_compact_certificate.md; lean/Taeyoung/Examples/Graph171.lean
172	Elf_	01, 03, 04, 05, 12, 15, 23, 25, 34	notes/turan_local_and_high_density_negative_tests.tex; lean/Taeyoung/Examples/Graph172.lean
173	ElMg	01, 03, 05, 12, 23, 24, 25, 34, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/verify_negative_witnesses.py; lean/Taeyoung/Examples/Graph173.lean
174	EtTg	01, 02, 03, 14, 15, 23, 25, 34, 45	notes/atlas174_compact_certificate.md; lean/Taeyoung/Examples/Graph174.lean
177	En{G	01, 03, 04, 12, 13, 14, 23, 24, 34, 45	notes/six_verified_base_cones.tex; lean/Taeyoung/Examples/Graph177.lean
178	En}?	01, 03, 04, 05, 12, 13, 14, 23, 24, 34	notes/atlas178_half_degree_weighted_k4.tex; lean/Taeyoung/Examples/Graph178.lean
179	E_~w	01, 04, 05, 14, 15, 24, 25, 34, 35, 45	notes/six_verified_base_cones.tex; lean/Taeyoung/Examples/Graph179.lean
180	EjtW	01, 04, 12, 13, 14, 15, 23, 34, 35, 45	notes/two_root_triangle_leaf_cones.tex; lean/Taeyoung/Examples/Graph180.lean
181	E^mG	02, 03, 04, 05, 12, 13, 23, 24, 34, 45	notes/atlas181_compact_certificate.md; lean/Taeyoung/Examples/Graph181.lean
182	E^Mg	02, 03, 05, 12, 13, 23, 24, 25, 34, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/verify_negative_witnesses.py; lean/Taeyoung/Examples/Graph182.lean

Atlas	graph6	edge list	source
183	EjvG	01, 04, 05, 12, 13, 14, 15, 23, 34, 45	notes/six_verified_base_cones.tex; lean/ Taeyoung/Examples/Graph183.lean
184	Elfo	01, 03, 04, 05, 12, 15, 23, 25, 34, 35	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph184.lean
185	Exv_	01, 02, 04, 05, 12, 14, 15, 23, 25, 34	notes/atlas185_compact_certificate.md; lean/Taeyoung/Examples/Graph185.lean
186	ErXw	01, 02, 13, 14, 15, 23, 24, 25, 35, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph186.lean
187	Ehfw	01, 04, 05, 12, 15, 23, 25, 34, 35, 45	notes/cliقة_oddcycle_join.tex; lean/Taeyoung/ Examples/Graph187.lean
188	EzNG	01, 02, 05, 12, 13, 15, 23, 24, 34, 45	notes/atlas188_compact_certificate.md; lean/Taeyoung/Examples/Graph188.lean
189	E^NG	02, 03, 05, 12, 13, 15, 23, 24, 34, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph189.lean
190	EyUw	01, 02, 05, 12, 13, 14, 25, 34, 35, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph190.lean
191	E~ ?	01, 02, 03, 04, 12, 13, 14, 15, 23, 24, 34	notes/cliقة_common_leaf_extensions.tex; lean/ Taeyoung/Examples/Graph191.lean
192	E~Xo	01, 02, 03, 12, 13, 14, 15, 23, 24, 25, 35	notes/six_verified_base_cones.tex; lean/ Taeyoung/Examples/Graph192.lean
193	En}G	01, 03, 04, 05, 12, 13, 14, 23, 24, 34, 45	notes/page_rooted_triangle_book_leaves.tex; lean/Taeyoung/Examples/Graph193.lean
194	E~wW	01, 02, 03, 04, 12, 13, 14, 23, 24, 35, 45	notes/atlas194_compact_certificate.md; lean/Taeyoung/Examples/Graph194.lean
195	EyVw	01, 02, 05, 12, 13, 14, 15, 25, 34, 35, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph195.lean
196	ER~g	02, 04, 05, 13, 14, 15, 23, 24, 25, 34, 45	notes/open_conditional_implications.tex; lean/ Taeyoung/Examples/Graph196.lean
197	E}^G	01, 02, 03, 05, 12, 13, 14, 15, 24, 34, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph197.lean

Atlas	graph6	edge list	source
198	Ep~o	01, 02, 04, 05, 14, 15, 23, 24, 25, 34, 35	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph198.lean
199	El^g	01, 03, 05, 12, 14, 15, 23, 24, 25, 34, 45	notes/atlas199_compact_certificate.md; lean/Taeyoung/Examples/Graph199.lean
200	E~{W	01, 02, 03, 04, 12, 13, 14, 23, 24, 34, 35, 45	notes/six_verified_base_cones.tex; lean/ Taeyoung/Examples/Graph200.lean
201	E~z_	01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph201.lean
202	Ep~w	01, 02, 04, 05, 14, 15, 23, 24, 25, 34, 35, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph202.lean
203	E~^G	01, 02, 03, 05, 12, 13, 14, 15, 23, 24, 34, 45	notes/atlas203_compact_certificate.md; lean/Taeyoung/Examples/Graph203.lean
204	EznW	01, 02, 04, 05, 12, 13, 15, 23, 24, 34, 35, 45	lean/Taeyoung/Methods/Negative/Tensor.lean; history/paper_independent/math_paper/scripts/ verify_negative_witnesses.py; lean/Taeyoung/ Examples/Graph204.lean
205	E~~G	01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 34, 45	notes/six_verified_base_cones.tex; lean/ Taeyoung/Examples/Graph205.lean
206	E~nW	01, 02, 03, 04, 05, 12, 13, 15, 23, 24, 34, 35, 45	notes/turan_local_and_high_density_negative_ tests.tex; lean/Taeyoung/Examples/Graph206.lean
207	Ez~w	01, 02, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph207.lean
208	E~~w	01, 02, 03, 04, 05, 12, 13, 14, 15, 23, 24, 25, 34, 35, 45	notes/pure_chordal.tex; lean/Taeyoung/Examples/ Graph208.lean

C Exact negative evaluations

Table 3: Exact counterexamples for the 23 negative rows.

[illegible]

D Retained Lean build measurements

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Table 4: Measured sequential Lean builds for the twenty compact-certificate rows. Time includes the fresh row build and the installed-source check when reported.

Atlas	source files	time (min)	peak memory (GiB)
118	89	25.7	9.09
122	89	25.9	9.09
124	89	25.8	9.09
126	100	45.3	9.73
127	97	43.6	11.90
130	100	34.6	9.21
147	97	43.8	11.84
151	97	46.2	11.83
153	98	40.9	11.90
157	89	27.0	9.20
168	97	44.7	11.84
169	89	26.9	9.23
171	98	42.4	11.88
174	98	41.0	11.91
181	89	26.9	9.23
185	89	28.6	9.23
188	98	41.0	11.90
194	89	26.8	9.21
199	89	26.8	9.23
203	100	37.5	9.26
Total	–	701.5	–